OCR
Oxford Cambridge and RSA

# Tuesday 4 June 2019 – Afternoon

## A Level Chemistry A

**H432/01** Periodic table, elements and physical chemistry

**Time allowed: 2 hours 15 minutes**

**You must have:**
- the Data Sheet for Chemistry A (sent with general stationery)

**You may use:**
- a scientific or graphical calculator

*H432/01*

Please write clearly in black ink. **Do not write in the barcodes.**

Centre number | | | | | | Candidate number | | | | |

First name(s) _____

Last name _____

**INSTRUCTIONS**
- Use black ink. You may use an HB pencil for graphs and diagrams.
- Answer **all** the questions.
- Where appropriate, your answers should be supported with working. Marks may be given for a correct method even if the answer is incorrect.
- Write your answer to each question in the space provided. If additional space is required, use the lined page(s) at the end of this booklet. The question number(s) must be clearly shown.

**INFORMATION**
- The total mark for this paper is **100**.
- The marks for each question are shown in brackets **[ ]**.
- Quality of extended responses will be assessed in questions marked with an asterisk (*).
- This document consists of **32** pages.

© OCR 2019 [601/5255/2]
DC (ST/JG) 174257/7

OCR is an exempt Charity

## SECTION A

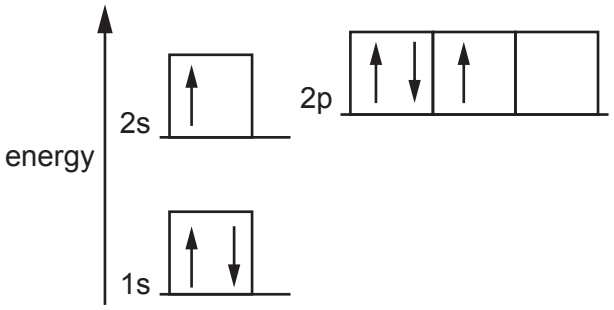
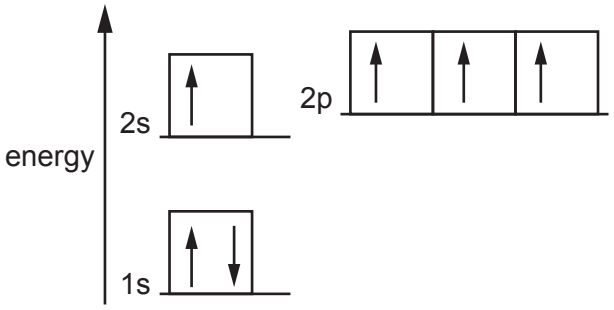
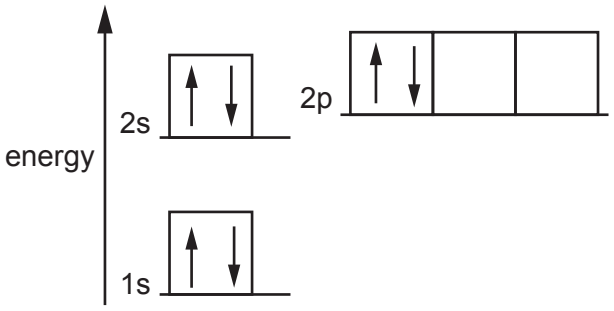
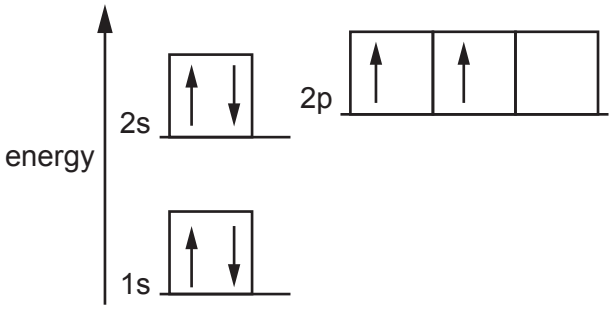
**You should spend a maximum of 20 minutes on this section.**

**Write your answer to each question in the box provided.**

Answer **all** the questions.

**1** In the diagrams below, each box represents an orbital and each electron is shown as an arrow.

Which diagram shows the correct arrangement of electrons in an atom of carbon?

| | |
|---|---|
| **A** |  |
| **B** |  |
| **C** |  |
| **D** |  |

Your answer D **[1]**

© OCR 2019

**2** Which statement about the reactions of halogens with halide ions is correct?

**A** $I_2(aq)$ can oxidise $Br^-(aq)$.

**B** $Cl_2(aq)$ can reduce $Br^-(aq)$.

**C** $Br^-(aq)$ can reduce $Cl_2(aq)$.

**D** $Cl^-(aq)$ can oxidise $I_2(aq)$.

F
Cl
Br
I

Your answer [A] **[1]**

**3** One molecule of a gas has a mass of $2.658 \times 10^{-23}$ g.

What is a possible formula of the gas?

$6.022 \times 10^{-23} = 1$

**A** $CH_4$

**B** $O_2$

**C** $SO_2$

**D** $SO_3$

Your answer [ ] **[1]**

**4** In the laboratory, acid spills can be cleaned up and made safe by spreading anhydrous sodium carbonate over the spill to neutralise the acid.

A student accidentally spills 50.0 $cm^3$ of 2.00 mol $dm^{-3}$ $HCl(aq)$ on the bench.

What is the minimum mass of anhydrous sodium carbonate required to neutralise the acid?

$Na_2CO_3$

$(23 \times 2) + 12 + 3(16)$

$46 + 12 + 48$

$106$

**A** 4.15 g

**B** 5.30 g

**C** 8.30 g

**D** 10.6 g

Your answer [ ] **[1]**

$moles = \frac{2}{0.05} = 40$

4 moles of $Na_2CO_3$

moles × RFM = Mass

© OCR 2019

**5** What is the oxidation number of N in $Mg(NO_2)_2 \cdot 3H_2O$?

**A** +2

**B** +3

**C** +4

**D** +5

Your answer [ ] **[1]**

**6** Which reaction is a redox reaction?

**A** $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$

**B** $NaNO_2 + HCl \rightarrow NaCl + HNO_2$

**C** $CaSO_3 + 2HCl \rightarrow CaCl_2 + H_2O + SO_2$

**D** $3CuO + 2NH_3 \rightarrow 3Cu + 3H_2O + N_2$

Your answer [ ] **[1]**

**7** Which set of elements in the solid state contain a simple molecular lattice, a giant covalent lattice and a giant metallic lattice?

**A** S, Si, Al

**B** P, Si, C

**C** S, P, Si

**D** Mg, P, S

Your answer [ ] **[1]**

© OCR 2019

**8** 3.528 g of a Group 2 metal, **M**, is reacted with an excess of chlorine.
The reaction forms 9.775 g of a chloride.

What is metal **M**?

**A** magnesium

**B** calcium

**C** strontium

**D** barium

Your answer ☐ **[1]**

**9** Which statement is **not** correct for Group 2 hydroxides?

**A** $Mg(OH)_2$ can be used to treat indigestion.

**B** $Ca(OH)_2$ is used in agriculture to neutralise alkaline soils.

**C** The anion in $Sr(OH)_2$ contains 10 electrons.

**D** $Ba(OH)_2$ is a product from the reaction of barium and water.

Your answer ☐ **[1]**

**10** Radical reactions are responsible for the catalysed breakdown of the ozone layer.

The overall equation is shown below.

$2O_3 \rightarrow 3O_2 \qquad \Delta_r H = -284\,kJ\,mol^{-1}$

The molar gas volume in the ozone layer is approximately $2.5\,m^3\,mol^{-1}$.

What is the energy released, in kJ, during the breakdown of $1.0\,m^3$ of ozone in the ozone layer?

**A** 56.8

**B** 113.6

**C** 355

**D** 710

Your answer ☐ **[1]**

© OCR 2019

**11** A graph of $\ln k$ against $\frac{1}{T}$ ($T$ in K) for a reaction has a gradient with the numerical value of −4420.

What is the activation energy, in $kJ\,mol^{-1}$, for this reaction?

**A** −532

**B** −36.7

**C** +36.7

**D** $+5.32 \times 10^5$

Your answer ☐ **[1]**

**12** The equation shows the dissociation of the acid $H_3AsO_4$ in water.

$H_3AsO_4 + H_2O \rightleftharpoons H_2AsO_4^- + H_3O^+$

Which pair is a conjugate acid–base pair?

**A** $H_3AsO_4$ and $H_2O$

**B** $H_2AsO_4^-$ and $H_3O^+$

**C** $H_3AsO_4$ and $H_3O^+$

**D** $H_3O^+$ and $H_2O$

Your answer ☐ **[1]**

**13** What is the number of stereoisomers that $Ni(H_2NCH_2CH_2NH_2)_2Cl_2$ can form?

**A** 2

**B** 3

**C** 4

**D** 6

Your answer ☐ **[1]**

© OCR 2019

**14** Which property/properties is/are correct for a transition element?

1 The element has atoms with a partially filled d sub-shell.

2 The existence of more than one oxidation state in its compounds.

3 The formation of coloured ions.

**A** 1, 2 and 3

**B** Only 1 and 2

**C** Only 2 and 3

**D** Only 1

Your answer ☐ **[1]**

**15** Four redox systems relevant to hydrogen–oxygen fuel cells are shown below.

| | $E^{\ominus}$ / V |
|---|---|
| $H_2O(l) + e^- \rightleftharpoons OH^-(aq) + \frac{1}{2}H_2(g)$ | −0.83 |
| $H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$ | 0.00 |
| $\frac{1}{2}O_2(g) + H_2O(l) + 2e^- \rightleftharpoons 2OH^-(aq)$ | +0.40 |
| $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O(l)$ | +1.23 |

Which statement(s) is/are correct for an alkaline hydrogen–oxygen fuel cell?

1 The reaction at the positive electrode is: $\frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O(l)$.

2 The overall cell reaction is: $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(l)$.

3 The cell potential is 1.23 V.

**A** 1, 2 and 3

**B** Only 1 and 2

**C** Only 2 and 3

**D** Only 1

Your answer ☐ **[1]**

© OCR 2019

**BLANK PAGE**

**PLEASE DO NOT WRITE ON THIS PAGE**

© OCR 2019

## SECTION B

Answer **all** the questions.

**16** Sir Humphry Davy discovered several elements including sodium, potassium, magnesium, calcium and strontium.

**(a)** Explain which block in the Periodic Table sodium and magnesium belong to.

.....

..... **[1]**

**(b)** A sample of magnesium, $A_r$ = 24.305, is found to consist of three isotopes. The accurate relative isotopic masses and % abundances of two of the isotopes are shown in the table.

| Isotope | Relative isotopic mass | % abundance |
|---|---|---|
| $^{24}$Mg | 23.985 | 78.99% |
| $^{25}$Mg | 24.986 | 10.00% |

Determine the relative isotopic mass of the third isotope of magnesium in the sample.

Give your answer to **5** significant figures.

relative isotopic mass = ..... **[2]**

**(c)** A student adds an excess of calcium oxide to water in a test tube.
In a separate test tube, the student adds an excess of strontium oxide to water.

**(i)** Write the equation for the reaction of calcium oxide with water.

State symbols are **not** required.

..... **[1]**

**(ii)** Suggest the approximate pH of the two solutions formed in the test tubes.

pH with calcium oxide .....

pH with strontium oxide.....
**[1]**

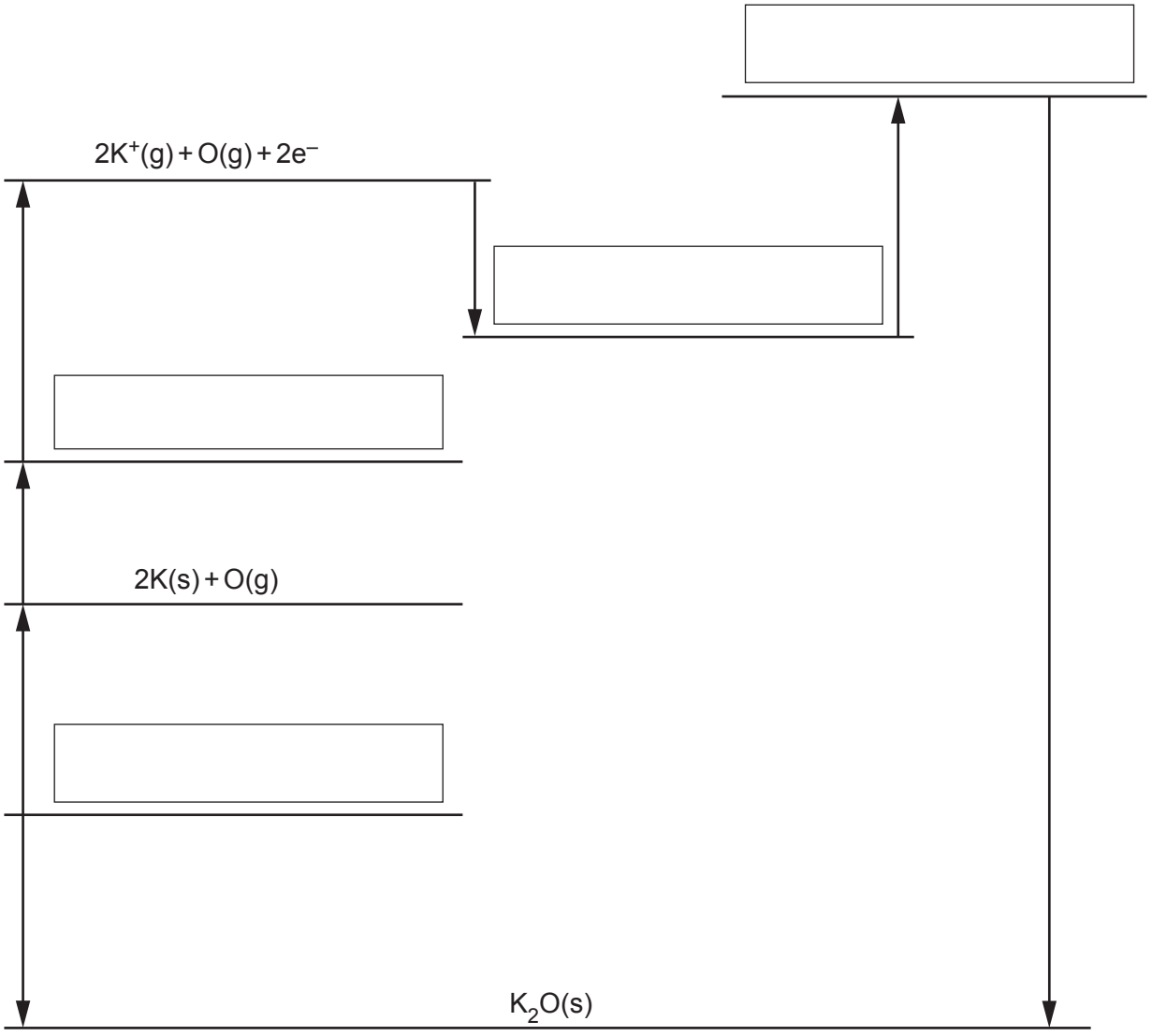
© OCR 2019

**(d)** The table below shows enthalpy changes involving potassium, oxygen and potassium oxide, $K_2O$.

| | **Enthalpy change / kJ mol$^{-1}$** |
|---|---|
| formation of potassium oxide | –363 |
| 1st electron affinity of oxygen | –141 |
| 2nd electron affinity of oxygen | +790 |
| 1st ionisation energy of potassium | +419 |
| atomisation of oxygen | +249 |
| atomisation of potassium | +89 |

**(i)** The incomplete Born–Haber cycle below can be used to determine the lattice enthalpy of potassium oxide.

In the boxes, complete the species present in the cycle.
Include state symbols for the species.




**[4]**

© OCR 2019

**(ii)** Calculate the lattice enthalpy of potassium oxide.

lattice enthalpy = ..... $kJ\,mol^{-1}$ **[2]**

**(e)** A similar Born–Haber cycle to potassium oxide in **(d)** can be constructed for sodium oxide.

**(i)** The first ionisation energy of sodium is more endothermic than that of potassium.

Explain why.

.....

.....

.....

.....

..... **[2]**

**(ii)** The lattice enthalpy of sodium oxide is more exothermic than that of potassium oxide.

Explain why.

.....

.....

.....

.....

..... **[2]**

© OCR 2019

**17** Healthy human blood needs to be maintained at a pH of 7.40 for the body to function normally.

**(a)*** Carbonic acid, $H_2CO_3$, is a weak acid which, together with hydrogencarbonate ions, $HCO_3^-$, acts as a buffer to maintain the pH of blood.

The p$K_a$ value for the dissociation of carbonic acid is 6.38.

Explain, in terms of equilibrium, how the carbonic acid–hydrogencarbonate mixture acts as a buffer in the control of blood pH, and calculate the $[HCO_3^-]$ : $[H_2CO_3]$ ratio in healthy blood. **[6]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Additional answer space if required

.....

.....

.....

.....

.....

.....

© OCR 2019

**(b)** Red blood cells contain haemoglobin.

Explain using ligand substitutions:

- how haemoglobin transports oxygen around the body
- why carbon monoxide is toxic.

.....

.....

.....

.....

.....

.....

..... **[3]**

© OCR 2019

**18** This question is about reactions of ions and compounds of transition elements.

**(a)** A student carries out two experiments on a solution containing $[Cr(H_2O)_6]^{3+}(aq)$.

**Experiment 1**
The student adds an excess of aqueous ammonia to a solution containing $[Cr(H_2O)_6]^{3+}(aq)$ until a purple solution is formed.

**Experiment 2**
The student carries out the following reaction sequence.

**Step 1** NaOH(aq) is added slowly to a solution containing $[Cr(H_2O)_6]^{3+}(aq)$ in a boiling tube.
A grey–green precipitate forms.

**Step 2** An excess of NaOH(aq) is added to the boiling tube.
The precipitate dissolves and a green solution forms containing a 6 coordinate complex ion.

**Step 3** $H_2O_2$ is added to the mixture and the boiling tube is heated.
A yellow solution forms.

**Step 4** The solution in the boiling tube is acidified.
The solution now contains $Cr_2O_7^{2-}(aq)$.

**(i)** What is the formula of the complex ion in the purple solution that forms in **Experiment 1**?

..... **[1]**

**(ii)** Suggest an equation for the reaction in **Experiment 2**, **Step 1**.
Include state symbols.

..... **[1]**

**(iii)** Draw a 3-D diagram for the shape of the complex ion that forms in **Experiment 2**, **Step 2**.
Include the charge of the ion.

[ ] .....

**[2]**

**(iv)** What is the formula of the **ion** that causes the yellow colour in **Experiment 2**, **Step 3**?

..... **[1]**

**(v)** State the colour of the solution that forms in **Experiment 2**, **Step 4**.

..... **[1]**

© OCR 2019

**(b)** Vanadium ions have four common oxidation states. **Table 18.1** shows the colours of the ions in aqueous solution.

| Oxidation state of vanadium | Vanadium ion | Colour |
|---|---|---|
| +5 | $VO_2^+(aq)$ | yellow |
| +4 | $VO^{2+}(aq)$ | blue |
| +3 | $V^{3+}(aq)$ | green |
| +2 | $V^{2+}(aq)$ | violet |

**Table 18.1**

**(i)** Complete the electron configuration of a $V^{3+}$ ion.

$1s^2$ ..... **[1]**

**(ii)** The student adds excess iron to a solution containing $VO^{2+}(aq)$ ions, and observes that the colour of the solution changes from blue to green and then to violet.

Use the relevant standard electrode potentials shown in **Table 18.2** to explain these observations.

| | Redox system | | | $E^{\ominus}$/V |
|---|---|---|---|---|
| **1** | $V^{2+}(aq) + 2e^-$ | $\rightleftharpoons$ | $V(s)$ | −1.18 |
| **2** | $Fe^{2+}(aq) + 2e^-$ | $\rightleftharpoons$ | $Fe(s)$ | −0.44 |
| **3** | $V^{3+}(aq) + e^-$ | $\rightleftharpoons$ | $V^{2+}(aq)$ | −0.26 |
| **4** | $VO^{2+}(aq) + 2H^+ + e^-$ | $\rightleftharpoons$ | $V^{3+}(aq) + H_2O(l)$ | +0.34 |
| **5** | $Fe^{3+}(aq) + e^-$ | $\rightleftharpoons$ | $Fe^{2+}(aq)$ | +0.77 |
| **6** | $VO_2^+(aq) + 2H^+ + e^-$ | $\rightleftharpoons$ | $VO^{2+}(aq) + H_2O(l)$ | +1.00 |

**Table 18.2**

.....

.....

.....

.....

.....

..... **[3]**

**(iii)** Construct an equation for the **first** colour change from blue to green.

..... **[1]**

© OCR 2019

**(c)** Iron(II) gluconate, $C_{12}H_{22}FeO_{14}$, is the active ingredient in some brands of iron supplements.

A student carries out an experiment to determine the mass of iron(II) gluconate in one tablet of an iron supplement, using the method below.

**Stage 1** The student crushes two tablets and dissolves the powdered tablets in dilute sulfuric acid.

**Stage 2** The student makes up the solution from **Stage 1** to 250.0 $cm^3$ in a volumetric flask.

**Stage 3** The student then titrates 25.0 $cm^3$ portions of the solution obtained in **Stage 2** with 0.00200 mol $dm^{-3}$ potassium manganate(VII).

The student obtains a mean titre of 13.50 $cm^3$.

In this titration, 1 mol of manganate(VII) ions reacts with 5 mol of iron(II) ions.

**(i)** Explain why the student used 0.00200 mol $dm^{-3}$ potassium manganate(VII) solution for this titration, rather than the more usual concentration of 0.0200 mol $dm^{-3}$ used in manganate(VII) titrations.

.....

..... **[1]**

**(ii)** Use the student's results to determine the mass, in mg, of iron(II) gluconate in **one** tablet.

Give your answer to **3** significant figures.

mass of iron(II) gluconate in one tablet = ..... mg **[5]**

© OCR 2019

**(iii)** Some iron supplements contain iron(II) sulfate or iron(II) fumarate.

The information in **Table 18.3** is taken from the labels of two iron supplements, **A** and **B**.

| Iron supplement | Iron compound | Mass of iron compound in one tablet / mg |
|---|---|---|
| **A** | iron(II) sulfate, $FeSO_4$ | 180 |
| **B** | iron(II) fumarate, $C_4H_2FeO_4$ | 210 |

**Table 18.3**

Choose which iron supplement, **A** or **B**, would provide the greater mass of iron per tablet.

iron supplement: ..... **[1]**

© OCR 2019

**19** Sulfuric acid is an important chemical used to make detergents, fertilisers and dyes. It is manufactured in a multi-step process.

**(a)** In the first step of the manufacture of sulfuric acid, sulfur dioxide, $SO_2$, can be made from the combustion of hydrogen sulfide, $H_2S$, shown in **Reaction 1**.

$2H_2S(g) + 3O_2(g) \rightarrow 2SO_2(g) + 2H_2O(l)$ $\quad \Delta_r H = -1125\,kJ\,mol^{-1}$ **Reaction 1**

**(i)** Explain why the enthalpy change for **Reaction 1** has a negative value.

Use ideas about enthalpy changes associated with bond breaking and bond making.

.....

.....

..... **[1]**

**(ii)** Some standard entropy values are given below.

| **Substance** | $H_2S(g)$ | $O_2(g)$ | $SO_2(g)$ | $H_2O(l)$ |
|---|---|---|---|---|
| $S^{\ominus}$ **/ J K$^{-1}$ mol$^{-1}$** | 206 | 205 | 248 | 70 |

Using calculations, explain whether **Reaction 1** is feasible at 20 °C.

**Calculations**

**Explanation for feasible or non feasible** .....

..... **[4]**

© OCR 2019

**(iii)** Calculate the standard enthalpy change of formation, $\Delta_f H^{\ominus}$, of hydrogen sulfide using the enthalpy change for **Reaction 1**, and the standard enthalpy changes of combustion below.

| **Substance** | $\Delta_c H^{\ominus}$ **/ kJ mol$^{-1}$** |
|---|---|
| S(s) | −296.8 |
| $H_2(g)$ | −285.8 |

$2H_2S(g) + 3O_2(g) \rightarrow 2SO_2(g) + 2H_2O(l)$ $\Delta_r H = -1125\,kJ\,mol^{-1}$ **Reaction 1**

$\Delta_f H^{\ominus}$ of hydrogen sulfide = ..... kJ mol$^{-1}$ **[3]**

© OCR 2019

**(b)** The second step in the manufacture of sulfuric acid is the conversion of $SO_2$ into sulfur trioxide, $SO_3$, using **Equilibrium 1**.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197\,kJ\,mol^{-1}$ **Equilibrium 1**

An industrial chemist carries out some research into **Equilibrium 1**.

- The chemist fills a $10.2\,dm^3$ container with $SO_2(g)$ at RTP, and then adds 12.0 g of $O_2(g)$.
- The chemist adds the vanadium(V) oxide catalyst, and heats the mixture. The mixture is allowed to reach equilibrium at a pressure of 2.50 atm and a temperature of 1000 K.
- A sample of the equilibrium mixture is analysed, and found to contain 0.350 mol of $SO_3$.

**(i)** Write an expression for $K_p$ for **Equilibrium 1**.

Include the units.

units = ..... **[2]**

**(ii)** Determine the value of $K_p$ for **Equilibrium 1** at 1000 K.

Show all your working.

Give your answer to **3** significant figures.

$K_p$ = ..... **[5]**

© OCR 2019

**(iii)** The chemist repeats the experiment in **(b)** at a different temperature.

The chemist finds that the value of $K_p$ is greater than the answer to **(b)(ii)**.

Explain whether the temperature in the second experiment is higher or lower than 1000 K.

.....

.....

.....

..... **[2]**

**(iv)** Explain the significance of the expression: $K_p \gg 1$.

.....

..... **[1]**

© OCR 2019

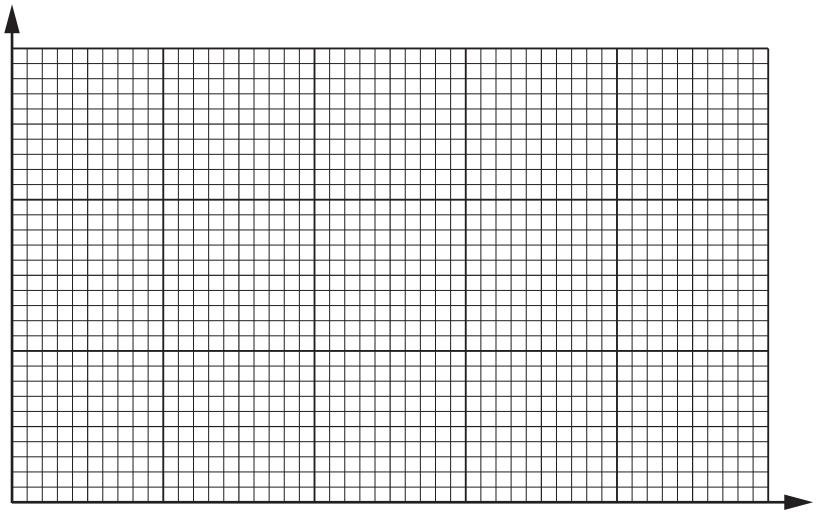
**(c)** Vanadium(V) oxide, $V_2O_5(s)$, is used as a catalyst in **equilibrium 1**.

$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ $\Delta H = -197\,kJ\,mol^{-1}$ **Equilibrium 1**

**(i)** Explain how the presence of $V_2O_5(s)$ increases the rate of reaction.

Include a labelled sketch of the Boltzmann distribution, on the grid below.

Label the axes.



.....

.....

.....

.....

..... **[4]**

**(ii)** Explain whether vanadium(V) oxide is acting as a homogeneous or heterogeneous catalyst.

.....

..... **[1]**

© OCR 2019

**BLANK PAGE**

**PLEASE DO NOT WRITE ON THIS PAGE**

© OCR 2019

**20** This question is about weak acids.

The $K_a$ values of three weak acids are shown in **Table 20.1**.

| Weak acid | $K_a$ / mol dm$^{-3}$ |
|---|---|
| iodic(V) acid, $HIO_3$(aq) | $1.78 \times 10^{-1}$ |
| propanoic acid, $C_2H_5COOH$(aq) | $1.35 \times 10^{-5}$ |
| hydrocyanic acid, HCN(aq) | $6.17 \times 10^{-10}$ |

**Table 20.1**

**(a)** Calculate the pH of 0.0800 mol dm$^{-3}$ $C_2H_5COOH$(aq).

Give your answer to **2** decimal places.

pH = ..... **[2]**

**(b)** A student adds a total of 45.0 cm$^3$ of 0.100 mol dm$^{-3}$ NaOH(aq) to 25.0 cm$^3$ of 0.0800 mol dm$^{-3}$ $C_2H_5COOH$(aq) and monitors the pH throughout.

**(i)** Show by calculation that 20.0 cm$^3$ of NaOH(aq) is required to reach the end point.

**[1]**

© OCR 2019

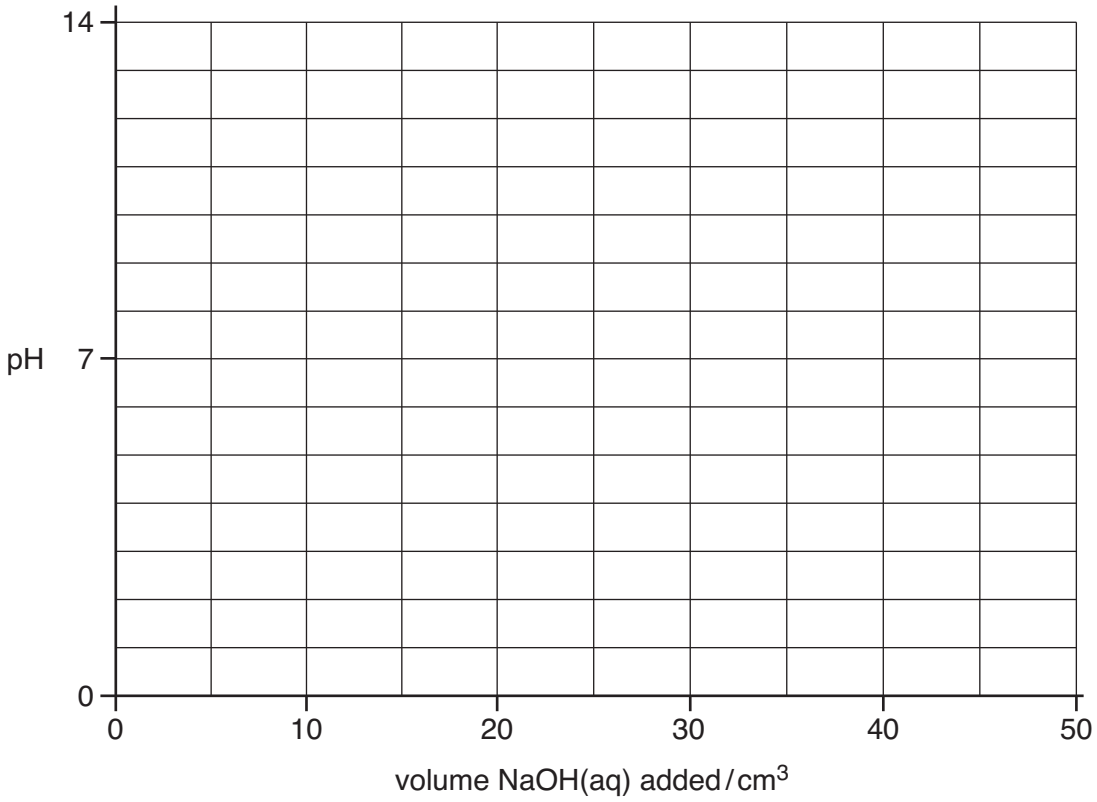
**(ii)** Calculate the pH of the final solution.

Give your answer to **2** decimal places.

pH = ..... **[4]**

© OCR 2019

**(iii)** On the axes below, sketch a pH curve for the pH changes during the addition of 45.0 $cm^3$ of 0.100 mol $dm^{-3}$ NaOH(aq) to 25.0 $cm^3$ of 0.0800 mol $dm^{-3}$ $C_2H_5COOH$(aq).




**[3]**

**(iv)** The student considers using the four indicators in **Table 20.2** for the titration.

| Indicator | pH range |
|---|---|
| Cresol red | 0.2 – 1.8 |
| Bromophenol blue | 3.0 – 4.6 |
| Cresol purple | 7.6 – 9.2 |
| Indigo carmine | 11.6 – 14.0 |

**Table 20.2**

Explain which indicator would be most suitable for the titration.

.....

.....

.....

.....

..... **[1]**

© OCR 2019

**(v)** The student repeats the experiment starting with 25.0 $cm^3$ of 0.0800 mol $dm^{-3}$ HCN(aq) and adding a total of 45.0 $cm^3$ of 0.100 mol $dm^{-3}$ NaOH(aq).

Predict **one** similarity and **one** difference between the pH curve with $C_2H_5COOH$(aq) and the pH curve with HCN(aq). Use the information in **Table 20.1**, and your answer to **(b)(iii)**.

Similarity .....

.....

Difference .....

.....

**[2]**

**(c)** The student calculates the pH of 0.0800 mol $dm^{-3}$ $HIO_3$(aq). The student assumes that the equilibrium concentration of $HIO_3$(aq) is the same as the initial concentration of $HIO_3$(aq).

The student measures the pH, and finds that the measured pH value is different from the calculated pH value.

Explain why the measured pH is different from the calculated pH.

.....

.....

.....

..... **[1]**

© OCR 2019

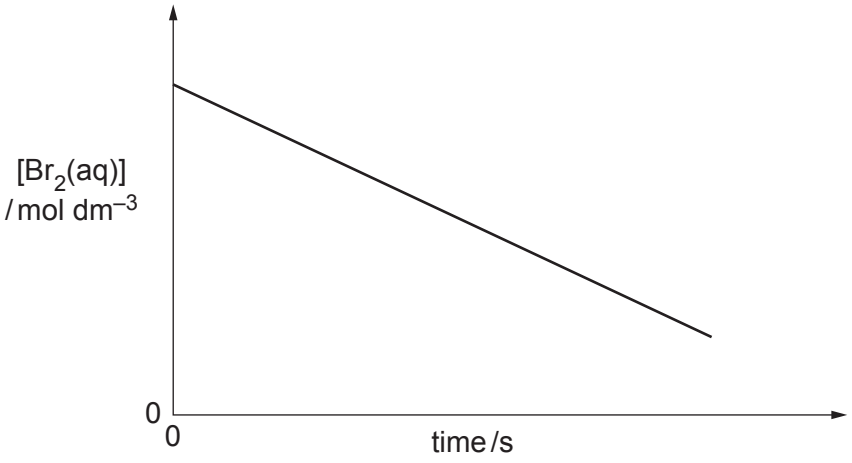
**21*** Three students carry out a rates investigation on the reaction between bromine and propanone in the presence of hydrochloric acid.

$CH_3COCH_3(aq) + Br_2(aq) \rightarrow CH_3COCH_2Br(aq) + HBr(aq)$

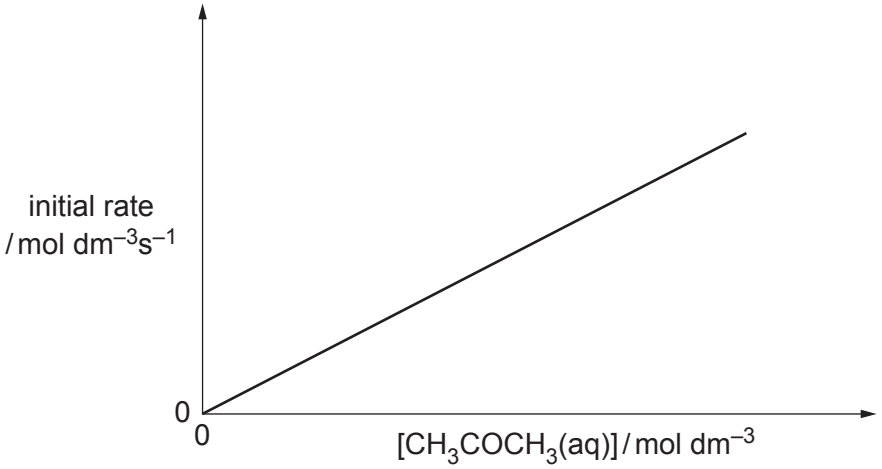
Each student investigates the effect of changing the concentration of one of the reactants whilst keeping the other concentrations constant.

Their results are shown below.

**Results of student 1**




**Results of student 2**




**Results of student 3**

| Experiment | $[Br_2(aq)]$ / mol dm$^{-3}$ | $[CH_3COCH_3(aq)]$ / mol dm$^{-3}$ | $[H^+(aq)]$ / mol dm$^{-3}$ | Initial rate / $10^{-5}$ mol dm$^{-3}$ s$^{-1}$ |
|---|---|---|---|---|
| **1** | 0.004 | 1.60 | 0.20 | 1.25 |
| **2** | 0.004 | 1.60 | 0.40 | 2.50 |

© OCR 2019

Explain how the reaction orders can be determined from the students' results, and determine the rate equation and rate constant. **[6]**

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Additional answer space if required

.....

.....

.....

.....

.....

.....

.....

.....

**END OF QUESTION PAPER**

© OCR 2019

## ADDITIONAL ANSWER SPACE

If additional space is required, you should use the following lined page(s). The question number(s) must be clearly shown in the margin(s).

© OCR 2019

© OCR 2019

OCR
Oxford Cambridge and RSA

**Copyright Information**

OCR is committed to seeking permission to reproduce all third-party content that it uses in its assessment materials. OCR has attempted to identify and contact all copyright holders whose work is used in this paper. To avoid the issue of disclosure of answer-related information to candidates, all copyright acknowledgements are reproduced in the OCR Copyright Acknowledgements Booklet. This is produced for each series of examinations and is freely available to download from our public website (www.ocr.org.uk) after the live examination series.

If OCR has unwittingly failed to correctly acknowledge or clear any third-party content in this assessment material, OCR will be happy to correct its mistake at the earliest possible opportunity.

For queries or further information please contact The OCR Copyright Team, The Triangle Building, Shaftesbury Road, Cambridge CB2 8EA.

OCR is part of the Cambridge Assessment Group; Cambridge Assessment is the brand name of University of Cambridge Local Examinations Syndicate (UCLES), which is itself a department of the University of Cambridge.

© OCR 2019